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A Review of Nitinol Shape Memory Alloys for Biomedical Applications: Advancements and Biocompatibility

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The equiatomic NiTi shape memory alloys (SMAs), known as nitinol, exhibit significant potential as a biomaterial for biomedical applications because of their distinctive properties, including the superelasticity and shape memory effect, which are superior to those of conventional materials. Due to their excellent biocompatibility, NiTi alloys are used in cardiovascular devices, orthopedic implants, surgical tools, and orthodontic devices. Despite these advantages, wear debris and nickel ion release challenges can compromise biocompatibility and provoke adverse cellular reactions, posing health risks. To enhance biocompatibility and prolong the implant lifespan, improvements in NiTi's mechanical (e.g., hardness values reported between 201 HV and 707 HV), tribological (coefficient of friction reduced to 0.31), and electrochemical properties are essential before human implantation. Moreover, NiTi alloys are generally considered non-cytotoxic if cell viability is > 70%. In addition, compared to conventional metallic biomaterials, NiTi alloys have a relatively low elastic modulus, ranging from 20 to 50 GPa in the martensitic phase and 40 to 90 GPa in the austenitic phase, which is much closer to that of natural bone (~30 GPa). This review paper summarizes current research and advancements in NiTi shape memory alloys, emphasizing their electrochemical and tribological properties. It also briefly addresses the biocompatibility issues associated with NiTi alloys and the strategies researchers employ to solve them.

INTRODUCTION

The selection of materials and manufacturing processes for biomedical implants is crucial when considering the exact locations and mobility conditions of human body structures such as joints and bones.[1] Enhancing the hardness, corrosion behavior, and wear properties of implants may increase their lifespan.[1]

Titanium (Ti) and its alloys are used as implant materials because of their exceptional properties.[2] Despite being utilized in a wide range of applications, conventional titanium alloys still exhibit several drawbacks that limit their long-term success. One major concern is their higher Young's modulus compared with bone, which can lead to bone atrophy and stress shielding in joints, leading to the failure of an implant.[3] Moreover, common alloys like Ti-6Al-4V can release potentially harmful metals like aluminum and vanadium, which can cause adverse biological reactions.[4,5] Additionally, the use of Ti alloys may be limited by certain characteristics, such as their low fatigue strength and low wear resistance, especially when they come

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into contact with hard components under cycling loading.[1,6] In vivo osseointegration may be slowed down by the bioinert surfaces and poor wear and corrosion behavior of Ti alloys.[1] The inherent limitations cause a great deal of pain and suffering for patients, and they often lead to issues like implant breakage because of insufficient mechanical properties and non-healing because of poor bone integration.[4] To address these problems associated with titanium alloys, research has focused on adding alloying elements to improve their properties.[7–10] Also, developing advanced preparation methods can improve microstructure and thus enhance performance within the human body,[11–13] in addition to using smart alloys that have unique qualities, such as shape memory.

Shape memory alloys (SMAs) are highly regarded as actuators in biomedical applications because of their exceptional biocompatibility, the unique ability to return to their original form when heated (shape memory effect) and their capacity to exhibit maximal recovery strain under mechanical tension (pseudoelasticity).[14] These alloys outperform other shape-memory polymers and hybrid materials because of their superior mechanical qualities and high actuation stress.[14] SMAs find utility as functional materials across various engineering applications, including biomedical implants, smart constructions, sensors and actuators, and aeronautical components.[15–17]

Among these, the nickel-titanium alloy (NiTi), commonly known as nitinol, has garnered the greatest interest because of its special qualities.[18–21] Beyond its shape memory capability, NiTi alloys exhibit other advantageous properties such as strength, ductility, low stiffness, superelasticity, and high resistance to wear and corrosion, coupled with excellent biocompatibility.[22–29]

These attributes significantly benefit cardiovascular devices, orthopedic implants and surgical equipment.[25,26,30,31] Common nitinol compositions include Ti (50%)-Ni (50%), Ti (45%)-Ni (55%), Ti (55%)-Ni (45%), and their derivatives.[22] Compared to other alloys, the material possesses a fine grain size and low anisotropy.[22] A critical feature of these materials is their high biocompatibility, which is defined as the ability to perform with an appropriate host response in a specific application.[32] This biocompatibility is mainly determined by the host reactions the material induces and its degradation in the bodily environment.[32]

Nitinol's exceptional corrosion resistance is attributed to the formation of stable, continuous, and highly adhesive protective oxide films on its surface.[33,34] These beneficial oxide layers naturally develop when freshly formed metal surfaces are exposed to air and/or moisture because titanium is highly reactive and has a strong affinity for oxygen.[33] During the implantation phase, the biological system will abrasively affect the implant and induce contact shear stress.[35] For example, aggressive ions, including chlorides, hydrocarbonates, and sulfates, as well as the high acidity of some bodily fluids and serum proteins, erode the NiTi implant's surface.[35–37] Because body fluids contain large quantities of Cl^- ions, NiTi implants are susceptible to pitting and crevice corrosion.[35] The material's pH is 7.35 at the time of implantation.[35] Pitting corrosion is generated by halogen materials such as iodide ions, bromide, and chloride and is distinguished by the local shape of the pits on the material's surface.[35] On the other hand, crevice corrosion is linked to elevated pH levels.[35] These localized corrosion processes may cause the oxide layer to deteriorate, exposing nickel-rich phases that are more susceptible to corrosion and causing nickel ions to be released.[36] This release can decrease the mechanical performance of implants and may trigger adverse cellular reactions.[38,39]

NiTi alloys exhibit superior wear resistant compared to traditional wear-resistant materials like steels and nickel- and cobalt-based tribo-alloys.[40–42] However, wear debris produced by subpar tribological characteristics or the discharge of nickel ions can compromise biocompatibility. Therefore, enhancing NiTi alloys' resistance to corrosion and wear is crucial for implant applications.

The corrosion and wear resistance of NiTi alloys can be significantly improved through various strategies, including surface modification (such as coatings and laser treatment), microstructural refinement, and heat treatments.[43–49] In addition, controlling preparation parameters such as grinding time in the mechanical alloy method helps to optimize the microstructure and reduce flaws, which lowers the rate of corrosion and improves wear performance.[25,26]

Although several studies have reviewed and examined the mechanical properties, corrosion resistance, tribological performance, and biocompatibility of nitinol (NiTi alloys) separately, comprehensive reviews that integrate these four aspects are uncommon. The purpose of this review is to close this gap by presenting a comprehensive analysis of NiTi as a shape memory alloy, examining its characteristics, applications, and production methods. Additionally, it explores experimental results and research advancements, elucidating NiTi's performance under concurrent mechanical and electrochemical loads in corrosive and tribological settings. The study also discusses its biocompatibility, presenting an overview of its potential for biomedical applications. Moreover, this review highlights the current challenges and future perspectives to enhance the performance of these alloys.

AN OVERVIEW OF NITINOL

NiTi is a group of titanium-based intermetallic compounds containing almost equal quantities of nickel (Ni) and titanium (Ti).[50,51] This alloy is usually referred to as "nitinol" because it was

discovered in 1963 by William J. Buehler and his colleagues at the Naval Ordnance Laboratory;[52,53] the letters "niti" and "nol" refer to nickel-titanium and Naval Ordnance Laboratory, respectively.[50,54,55]

NiTi's atomic structure is a three-dimensional symmetric grid, with four titanium atoms surrounding each nickel atom[56] (refer to Fig. 1). These atoms are bound together by atomic interactions to form a special crystal structure that may show a phase change.[56] The first phase is a stronger, higher-temperature condition called the austenite phase.[56,57] The second phase, known as the martensite phase, is weak and unstable, and it occurs at low temperatures. The two-phase transition gives NiTi alloys their distinct physicochemical characteristics.[56,57]

Despite the rapid development of multifunctional biomaterials for enhanced bioactivity, drug delivery, and tissue regeneration, NiTi SMAs remain essential in biomedical engineering because of their beneficial properties:

- **Shape memory effect (SME)** results from a material changing from one crystal form to another as it is cooled or heated over a range of typical transformation temperatures.[58] In NiTi alloys, the monoclinic crystal phase (martensite) replaces the ordered cubic crystal structure (austenite),[58] as illustrated in Fig. 2.
- **Superelasticity (pseudoelasticity):** superelasticity, also known as pseudoelasticity, is the extremely high elastic behavior of the alloy under stress.[59] For instance, a superelastic NiTi wire may typically produce reversible strains of up to 10% elongation, whereas a steel wire can only accomplish 0.5% reversible strain.[59] When stress is applied to the alloy, it transitions from the austenitic phase to the martensitic phase, exhibiting superelastic behavior in the austenitic phase.[59]
- **Low Young's modulus (elastic modulus):** the NiTi alloy's elastic modulus, which is significantly closer to that of natural bone (about 30 GPa), ranges from 20 GPa to 50 GPa for martensite and 40 GPa to 90 GPa for austenite.[58] The low modulus is advantageous in the orthopedic field because it transfers the necessary stress to neighboring bone without causing bone resorption, thus avoiding the loosening of implants and giving implants a more extended service period to prevent revision surgery.[58]
- **Good biocompatibility and resistance to corrosion:** NiTi alloy has excellent corrosion resistance and biocompatibility, mainly because a passive dense titanium-oxide layer forms on its surface (TiO_2).[25,58]

These characteristics make it ideal for long-term implantation, especially in applications that call for longevity and mechanical adaptation in physiological settings.

The phase diagram of the NiTi alloy system is crucial for improving the properties of alloys during heat treatment (Fig. 3). It presents four phases of equilibrium: $NiTi_2$, NiTi, Ni_3Ti, and Ni_4Ti_3:[60–62]

- The NiTi phase has an ordered B2 (CsCl) type structure; at room temperature, its lattice constant is 0.3015 nm. The B2 phase is preserved upon quenching or slow cooling to ambient temperature. This phase is crucial for the martensitic transformation and the related shape memory effects.
- The $NiTi_2$ phase has a cubic shape with space group Fd3m. The unit cell has 96 atoms with a lattice constant of 1.132 nm.
- The Ni_3Ti phase has a hexagonal ordered structure of the DO24 type. The lattice constants are c = 0.83067 nm, a = 0.51010 nm, and c/a = 1.6284.

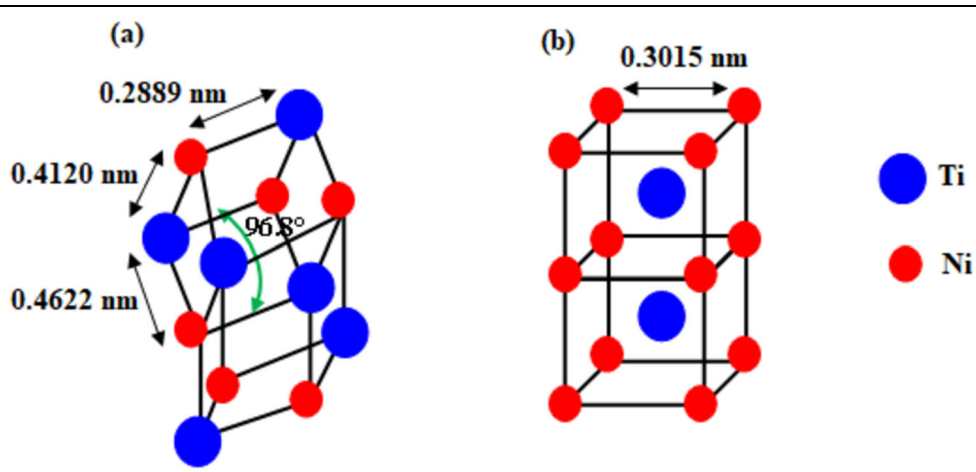

Fig. 1. Atomic structure of NiTi shape memory alloys: (a) martensite (monoclinic structure, a = 0.2889 nm, b = 0.4120 nm, c = 0.4622 nm, β = 96.8 °); (b) austenite (B2 cubic structure, a = 0.3015 nm).

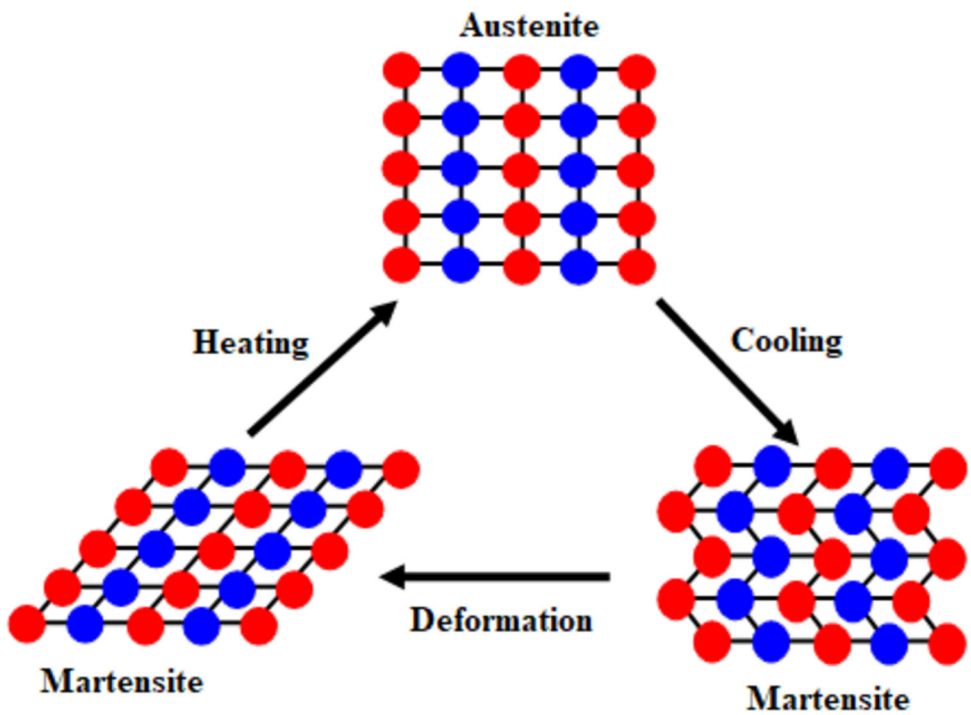

Fig. 2. Schematic illustration of the shape memory effect.

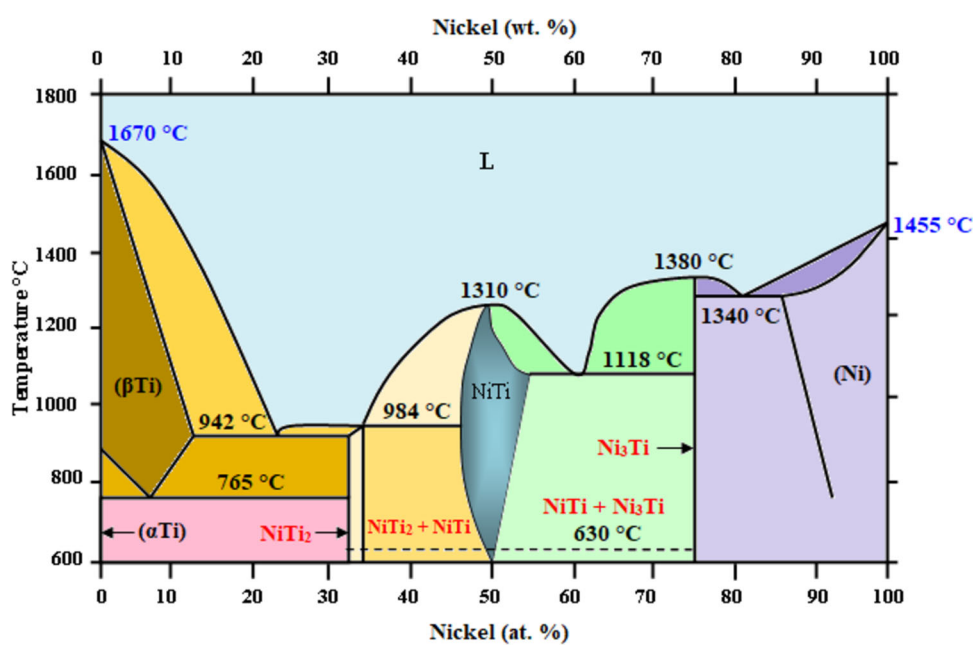

Fig. 3. NiTi alloy system's phase diagram.

- The Ni_4Ti_3 has a rhombohedral structure in space group R3. The lattice constant has a value of 0.670 nm.

As several authors have proved, controlling the martensitic transformation can significantly change NiTi's functional behavior. Various key factors affect the control of the martensitic transformation, for instance, chemical composition. Even a slight modification of 0.1 at.% in the composition of the alloy can significantly vary the onset and end temperatures of phase shifts, which in turn impact the potential application of NiTi in a range of smart structures in many different sectors and biomedicine.[63] In other words, the shape memory effect and superelasticity of the alloy are

determined by its functional temperatures, which can be influenced by the alloy's composition and microstructure.[63]

Furthermore, the thermodynamic and kinetic circumstances of martensitic transformation processes, which in turn affect the mechanical properties of NiTi alloy, are their microstructure, particularly dislocation substructures and Ni_4Ti_3 precipitation.[63,64] Additionally, the transformation sequence plays an important role in controlling these transformations. Aging treatment changes martensitic transformations, resulting in improved mechanical and corrosion behavior on NiTi alloys.[65,66] On the other hand, alloying elements like Nb not only change transition temperatures but also enhance yield stress and damping for applications that require superior mechanical performance.[67] The enhancements in mechanical characteristics brought about by regulating martensitic transformation result in a material that is highly biocompatible and resistant to wear and corrosion.[66,68]

MANUFACTURING OF NITINOL

Since the composition of titanium and nickel is crucial in determining the properties of NiTi alloy, manufacturing techniques have been developed to create high-quality NiTi alloy in tandem with the expansion of their use in biomedical devices.[69] Figure 4 below depicts NiTi synthesis procedures. The two metals usually have a ratio of 50/50 (at.%); any variation in this ratio modifies the superelastic and shape memory characteristics.[25,69] NiTi alloys are challenging to make and work with because of titanium's high spontaneity and extremely low machinability.[22] Traditional processing techniques, such as powder metallurgy and casting, are faced with several challenges in NiTi processing.

The following is a summary of these challenges:[15,22,70,71]

- Reaching a homogeneous and consistent composition.
- Creating complex geometries.
- Purchasing high-quality NiTi powder that is devoid of impurities and oxides.
- Inert atmosphere provision (to prevent oxidation).
- Preventing contaminants (which impact crack propagation and transition temperatures).
- Preventing defects and undesirable porosity (which lower load and starts crack nucleation and propagation).

Casting/Melting Processes

Due to the high titanium content, molten NiTi alloys are highly reactive and require treatment under vacuum.[72,73] The most often used method for

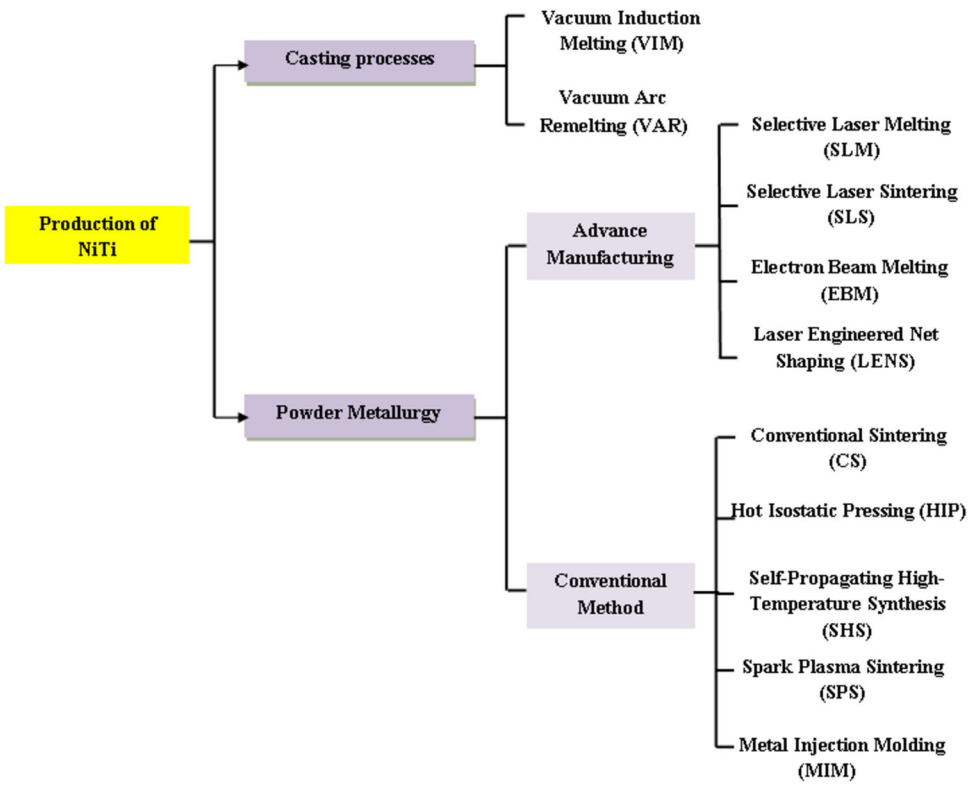

Fig. 4. Methods used to produce NiTi alloys.

producing NiTi alloys is casting.[73–76] In casting, molten material—typically metal—is poured into a mold cavity to make the final product.[77] After cooling, the molten substance hardens into the required shape, usually with the help of the mold.[77] Casting is typically a somewhat difficult process because of the intricate metallurgy involved in employing molten metal, even though the above describes a simple technique.[77] Two major categories can be used to classify casting procedures:[77]

- Expendable mold processes;
- Permanent mold processes.

The two most used melting techniques for the commercial manufacturing of NiTi alloys are vacuum induction melting (VIM) and vacuum arc remelting (VAR).[78,79]

Vacuum Induction Melting

The method most frequently employed for the commercial fabrication of NiTi alloys is vacuum induction melting.[79,80] The ingot exhibits greater chemical and microstructural homogeneity because of the superior churning of the melt caused by the electrodynamic forces in VIM.[79] Carbon has a strong affinity for titanium and is highly soluble in nickel.[79] Thus, the carbon contamination of the ingot from the graphite crucible is the main disadvantage of the VIM.[79]

Vacuum Arc Remelting

A secondary metallurgical technique called vacuum arc remelting is utilized to create ultraclean steel alloys as well as alloys based on nickel and titanium.[81] Through a DC arc, heat is delivered to a remelting electrode.[81] In VAR, a consumable or nonconsumable electrode is continuously remelted by an arc in a vacuum environment.[73] VAR melting produces extremely pure alloys, which enhance the structure and increase the cleanliness of VIM ingots.[73,82] To attain the needed homogeneity, multiple melts could be necessary because the entire ingot is not melted at once.[73,83] The costs of VIM and VAR are comparable.[73,83]

Powder Metallurgy Processes

Powder metallurgy (PM) is a metal-forming technique utilizing ceramic or metallic powders as raw materials.[84] These powders are sized and formed precisely before being sintered at a high temperature.[84] In this technology, a powder is typically defined as a particle < 200 μm.[85] In mass production, powders < 100 μm are commonly used; however, many particular PM-forming methods utilize powders < 20 μm.[85] Using a methodical approach, powder metallurgy turns metal particles into metallic objects.[86] The sintering process entails raising the temperature of the compressed metal powders below their melting point, enabling atomic diffusion and bonding and solidifying the material into a rigid structure.[86] This manufacturing procedure solves the problem of the liquid metal matrix and reinforces the material by not wetting it because the mixing is performed in a solid form.[86] Figure 5 displays the schematic diagram of the process for powder metallurgy.

To improve the uniformity of the powders, preblending techniques, including ball milling (BM), were used for mixing.[25,26,87] Carefully selecting the ball-to-powder weight ratio, milling speed, and milling time ensures improved manipulation of the mixed powder structure.[87] The advantages of this method over the conventional stirring casting technique include reduced manufacturing steps, higher material utilization, and improved dimensional precision.[84] Furthermore, PM technology may improve the microstructure, remove alloy segregation, and raise the supersaturation of alloying elements.[84]

Powder metallurgy has the potential to produce near-net-shape NiTi components because it can eliminate or minimize the need for expensive thermo-mechanical treatments, associated material losses, and machining.[16,84,88,89] However, it is well recognized that a significant rise in impurities, particularly carbon and oxygen, is inevitable and is a disadvantage of PM processing NiTi.[65,90] Precipitates associated with impurities alter the NiTi matrix' composition and could deteriorate its mechanical and functional qualities.[89,90]

PM techniques include conventional and additive manufacturing (AM) methods. Conventional PM techniques encompass conventional sintering, metal injection molding, spark plasma sintering, hot isostatic pressing, and self-propagating high-temperature synthesis.[89] However, additive manufacturing PM processes include selective laser melting, selective laser sintering, laser engineered net shaping, and electron beam melting.[89] Table I lists these procedures' advantages and disadvantages.

ADDITIVE MANUFACTURING OF NITINOL ALLOYS

Additive manufacturing is a novel technique that produces three-dimensional (3D) objects directly from digital models through an additive process that often entails the deposition of successive layers of polymers, ceramics, or metal components.[105]

In the realm of producing medical implants, both additive manufacturing and conventional powder technologies are important manufacturing pathways; nevertheless, their application areas are very different. Conventional PM offers the advantage of cost-effectiveness and the appealing potential to produce near-net-shape components, avoiding the challenging machining procedure needed to create bulk components through ingot metallurgy,[106] while AM gives precise control over the internal structure of porous materials, enabling the creation of tailored

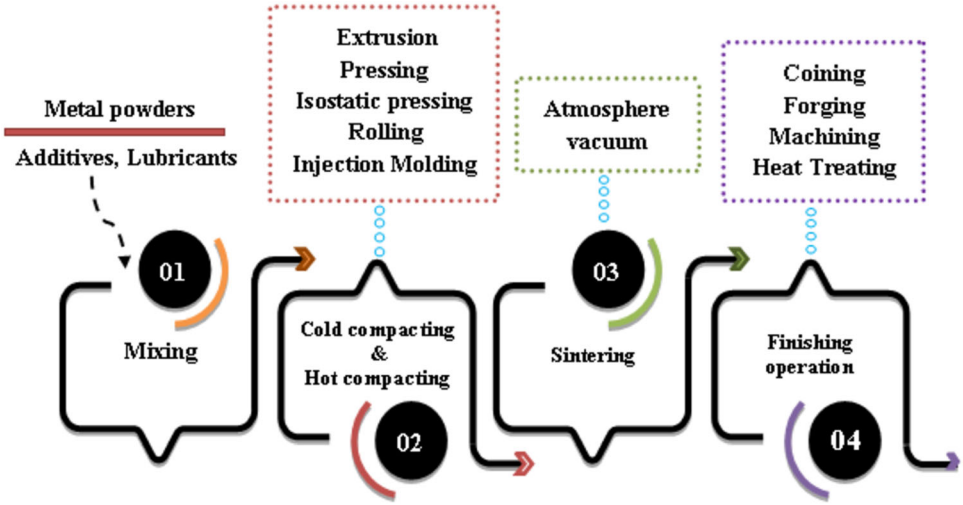

Fig. 5. Schematic of powder metallurgy process.

scaffolds for tissue engineering and regenerative medicine applications.[107–110] This technology has shown great ability to create complex geometries and lightweight structures, enhancing efficiency and producing customized orthopedic implants and prosthetics, as well as dental applications, enhancing patient comfort and outcomes.[107–110] AM successfully reduces the stress shielding effect caused by material modulus mismatch and promotes bone tissue growth and regeneration by carefully controlling the internal porosity structure of implants.[111] This also makes implants more biocompatible.[111]

Manufacturing methods greatly influence the structural, mechanical, and functional properties of materials. Furthermore, selecting the manufacturing process is crucial when considering both environmental (pollution and consumption) and economic (cost and time) factors.[112] For NiTi alloys, each preparation method has a direct effect on the microstructure and thus on its functional behavior. For example, the mechanical alloying process produces nanostructured materials because of extreme plastic deformation during milling. This leads to significant grain refinement and increased defect density, which in turn enhances its mechanical hardness and improves its resistance to corrosion and wear.[19,25] In contrast, additive manufacturing such as selective laser melting and laser powder bed fusion (LPBF) significantly affects the characteristics of NiTi alloys.[113,114] However, these processes often involve defects, including porosity, fractures, microstructure inhomogeneity, and impurity pickup, among others, which can affect the final product quality as well as the mechanical and thermal properties of the printed NiTi parts.[113] Post-AM heat treatments are therefore frequently required to optimize characteristics, particularly for biological applications.[41,113,114]

Recent initiatives have focused on improving AM procedures and using heat treatments to enhance NiTi-based alloys' functional properties for use in biomedicine, where microstructure control improves these alloys' qualities. Yuan et al.[115] investigated laser powder bed fusion of NiTiFe alloys from premixed powders and found that the elongation and fracture strength were both lowered to 23% and 1701 MPa, respectively, along with notable grain refinement. The evolution of mechanical properties is caused by incoherent oxide, low dislocation density, and large internal defects. Song et al.[116] examined the effect of heat treatment on Ni-45Ti-5Al-2Nb-1Mo alloys. The findings showed that solution treatment and subsequent aging cause precipitation from phases like Ni_2TiAl, which improves tensile strength and ductility. Witkowska et al.[117] focus on advancements in surface modification of NiTi alloys for orthopedic implants: focus on low-temperature glow discharge plasma oxidation techniques. This method produces uniform, nanocrystalline layers of titanium dioxide (TiO_2), titanium nitride (TiN), and nitrogen-doped TiO_2 layers, significantly enhancing corrosion resistance, reducing nickel ion release, and promoting osseointegration. Mustafa et al.[118] studied the porous NiTi dental implant fabricated by a metal injection molding. According to the study, in an animal model, porous NiTi implants were not expected to result in chronic inflammatory responses, acute systemic toxicity, or local sensitization. Additionally, porous NiTi demonstrated osseointegration comparable to that of Ti6Al4V, which is known to be biocompatible. Similarly, Li et al.[119] found that NiTi scaffolds made by selective laser melting have been tuned to enable osteoblast growth while exhibiting A_f temperatures close to body temperature (~37°C), good tensile strength of 612 MPa, and shape recovery rate of 96.94%.

Table I. Advantages and disadvantages of powder metallurgy NiTi SMAs production methods

Methods		Advantages	Disadvantages	Ref.
Conventional powder metallurgy processes	Conventional sintering	Elevated production rate - cheap - minimal need for additional processing - easy to use - good precision - less expensive than casting techniques - decreased wasting of materials	Prolonged warming periods - the sample's poor mechanical properties - limited pore shapes and sizes (at most 40% porosity)	89, 91–93
	Hot isostatic pressing	Effective use of materials -allows for a shorter diffusion time in solid states - high homogeneity - excellent control of the geometry and size of the pores - low sintering temperature - generates dense products having mechanical qualities that are isotropic - able to create vast or complicated products	Limitations on the form and size of pores (highest porosity of 40%) - intermetallic phases (Ni_3Ti, $NiTi_2$, and Ni_4Ti_3) exist - possible inert gas-induced porosities - low manufacturing rate	89, 91, 94
	Self-propagating high-temperature synthesis	Cheap - minimal energy consumption - notable porosity, up to 65%	Potential development of secondary phases due to unfinished reactions - ineffective intermetallic phase controls that result in embrittlement	89, 91, 95–98
	Spark plasma sintering	Easy to use - extremely accurate control - permits compressed powders to be sintered at a high density at comparatively low temperatures	It requires an expensive DC generator that only works with basic sample shapes	89, 91, 99, 100
	Metal injection molding	Exceptional geometric precision - ideal for products with a near-net shape - economical for large quantities of small-scale production - up to 98% density is possible	Expensive because of the costly tools -remaining pores - significant quantity of impurity phases	89, 91, 101
Additive manufacturing process	Selective laser sintering	Good control of shape - good porosity and homogeneity	It is costly due to expensive tooling, and there are also residual pores, several impurity stages, and part size limits	89, 91, 102
	Selective laser melting	Able to produce thick components (nitinol density of 97% is possible) - reduced beam distortion -the pieces do not have any distinct melt or binder phases	Costly - requires a lot of energy and is responsible for melting pool instability	89, 91, 97, 102
	Laser engineered net shaping	Capable of producing extremely pure components that are completely dense and have good microstructures	Inadequate surface polish -costly - slow -the materials must be powders	89, 91, 103
	Electron beam melting	Provides quick scan speeds - minimal waste of materials - appropriate for substances with a strong affinity for oxygen - high porosity - the mechanical characteristics of the final products are similar to those of casting methods - low oxygen and carbon concentration (10 times lower than VIM)	Inadequate surface finish -costly - inaccurate dimensionalization - only appropriate for metallic products	89, 104

It has been demonstrated that the microstructural characteristics that AM introduces into NiTi can have a major effect on its electrochemical performance. Grain refinement through rapid solidification during AM can increase mechanical hardness and corrosion resistance.[120] Several parameters of the AM process, such as laser power, scanning speed, layer thickness, and hatch spacing, significantly affect the mechanical and structural properties of the resulting NiTi alloys,[121] which in turn affect their electrochemical behavior. Soheil et al.[122] found that laser power and scanning speed have a major effect on texture and grain size. Sayed et al.[123] determined that hatch spacing is the most crucial variable for attaining the required texture after analyzing the impact of input parameters on grain solidification. These microstructural changes (grain refinement) have a direct impact on the materials' density, porosity, hardness, and surface roughness, all of which contribute to improved corrosion resistance.[25]

ELECTROCHEMICAL BEHAVIOR OF NITINOL

Corrosion resistance is one of the primary features of metallic materials used in the environment of the human body. The capacity of an implant to function in hostile bodily fluids is strongly tied to its performance. These liquids often contain several acids and a specific quantity of NaCl.[124] The pH is 7 under normal circumstances, but an immune system reaction, such as an infection or inflammation, may change it.[124] The implant component may lose its integrity in the case of corrosion, which would lead to failure.[124] Furthermore, the discharge of corrosion products may trigger harmful biological processes.[124]

Titanium is regarded as a stable metal with excellent corrosion resistance, unlike copper and stainless steel.[125] This good corrosion resistance is due to the development of a thin oxide layer on its surface, which separates the metal from its environment.[126] The thickness of passive films formed on these metals is 4 to 6 nm, which protects them from corrosion and slows down the release of titanium ions.[127]

The corrosion characteristics of NiTi alloys are worrisome when considering cytocompatibility. The release of Ni ions (Ni^{2+}) from the matrix may negatively affect biological applications[25] (Fig. 6). The rapid development of protective oxide layers on the surface of NiTi alloys is responsible for their excellent corrosion resistance and, consequently, their biocompatibility.[128] On the NiTi surface, TiO_2 usually forms the majority of oxides, with trace amounts of Ti_2O_3, TiO, Ni_2O_3, and NiO.[128] On the other hand, Ni^{2+} can still leak in physiological settings when subjected to extended service periods.[38,128] Nevertheless, precise Ti/Ni ratio control can enhance corrosion resistance by stabilizing the TiO_2 coating.[128]

TiO_2 is produced when titanium alloys are exposed to oxidizing environments.[128] It has no oxygen and is a conductor of excess electrons.[128] According to the electroneutrality condition, oxygen vacancies must be balanced by an equal number of negative charges. This parity can be achieved by incorporating O^{2-} ions into the lattice, ensuring that the entire crystal remains electro-neutral.[128] The disorder of the oxide influences the mobility of species during oxidation.[128] The concentration gradient of the oxide's vacancies acts as the driving force for diffusion; for example, in TiO_2, oxygen ions diffuse inward because the oxide/metal contact has the largest concentration of oxygen vacancies.[128] Beyond determining the site of scale growth, a disorder in oxides also affects the scale's capacity to seal fissures through oxide regeneration at high temperatures.[128,129] NiTi oxidation takes place as follows:

$$\mathrm{NiTi} + \mathrm{O_2} \rightarrow \mathrm{Ni_3Ti} + \mathrm{TiO_2} \rightarrow \mathrm{Ni_4Ti} + \mathrm{TiO_2} \rightarrow \mathrm{Ni} + \mathrm{TiO_2}$$

For NiTi alloys to gain broader acceptance as an implantable material, the surface's form and structure must be improved to stop Ni release.[130]

Electrochemical studies are crucial for understanding the corrosion behavior of materials. Three essential techniques that are frequently employed in this study are: potentiodynamic polarization (PD), open circuit potential (OCP) measurement, and electrochemical impedance spectroscopy (EIS). The OCP measurements can identify the development of passive layers on metals, which are crucial for corrosion resistance. By analyzing the polarization curves, researchers can determine important parameters such as the corrosion potential and current density, as well as corrosion rates. EIS is an effective method for assessing electrolyte-passivation layers' capacitance, resistance, and interfacial characteristics in electrochemical systems.

For biomedical applications, the electrochemical analysis is usually carried out in physiological media that replicate the human body, such as Ringer's, phosphate-buffered saline (PBS), 0.9% NaCl,[131] or Hank's solutions.[25] The electrochemical tests were performed using a potentiostat connected to a microcomputer, employing a three-electrode cell configuration (Fig. 7):

- Working electrode: NiTi alloy specimens.
- Reference electrode: saturated calomel electrode (SCE).
- Auxiliary electrode: typically made of platinum (Pt).

Previous research on the electrochemical properties of NiTi SMAs has provided valuable insight into their functionality and uses. Buciumeanu et al.[132]

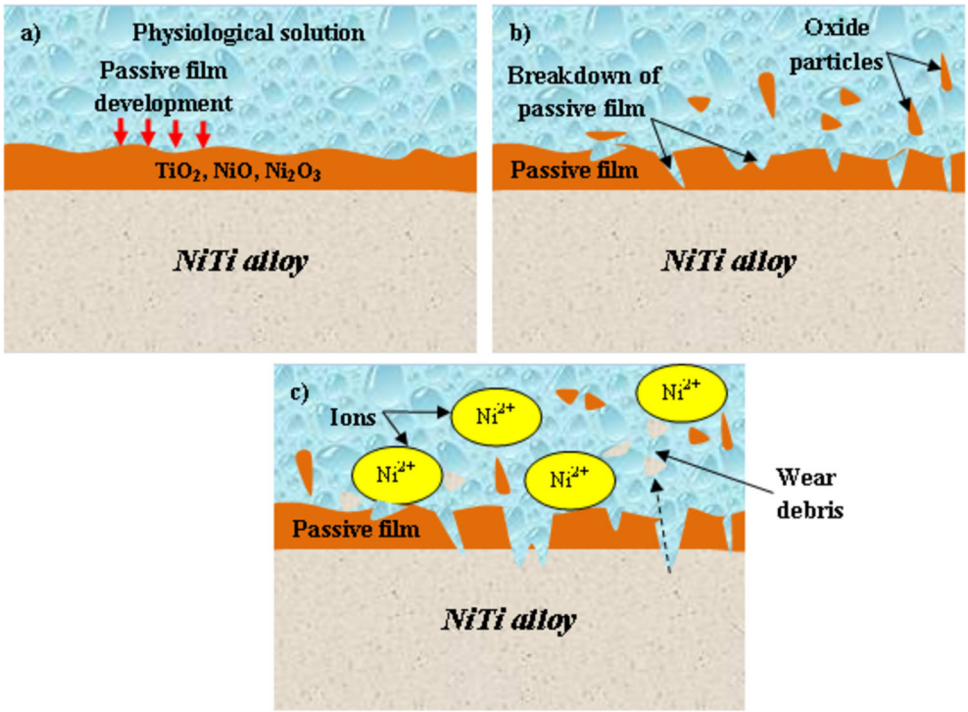

Fig. 6. Schematic illustration of the release of Ni ions from NiTi alloy: (a) formation of the passive layer; (b) breakdown of the passive layer; (c) Ni ion release.

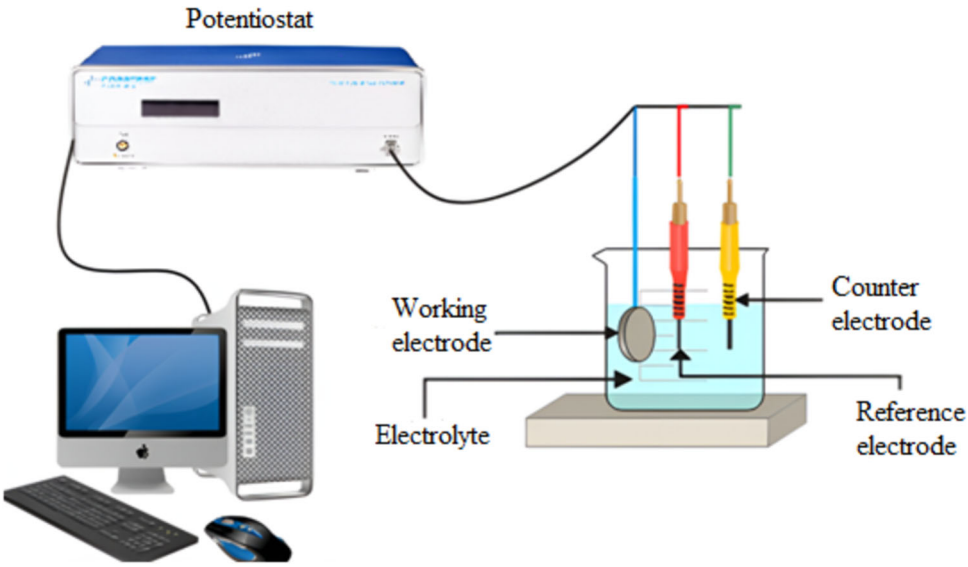

Fig. 7. Diagram of an electrochemical cell (three electrodes).

examined the combined effects of wear and corrosion in phosphate-buffered saline on as-printed NiTi SMA using laser beam-directed energy deposition (LB-DED) additive manufacturing methods, including laser-engineered net shaping (LENS). Using the same processing method (LB-DED and LENS), they also evaluated the tribocorrosion behavior of NiTi alloy with Ti-6Al-4V, the most popular titanium alloy for biomedical purposes. Their findings revealed that the NiTi alloy exhibits lower susceptibility to metallic ion discharge during its implant service life[132,133] than the Ti-6Al-4V alloy.[134]

In an in vivo study, Pértile et al.[135] evaluated the OCP of NiTi alloys during vascular procedures. Their results indicate that NiTi remains suitable for long-term implantation by maintaining a persistent thin surface film that guards against active degradation under physiological conditions. In addition, when corrosion and mechanical wear are coupled,

the NiTi alloy experiences greater overall wear in the simulated saliva than it would in the absence of a corrosive environment, as studied by Kosec et al.[136] Zhang et al.[137] used various manufacturing parameters, such as altered speed function and focus offset, to examine the corrosion behavior of NiTi SMA made by electron beam melting. The corrosion resistance of NiTi alloys was found to be impacted by the varying types and numbers of defects generated by variations in fabrication settings. When a few tiny pores were evenly distributed across the alloys, a greater corrosion resistance comparable to that of the wrought alloy was seen.

Still, the alloy with the most cracks showed the lowest corrosion resistance.[137] Several factors contributed to the formation of these cracks.[137] First, during the fabrication process, the material in the low-temperature area may limit the amount of material in the melting pool and the melting pool's boundary area, resulting in compressive stress. In the meantime, as the temperature rises, the material's yield strength will decrease in the heated area. When the compressive stress exceeds the decreased yield strength, the material will be in the plastic thermal compression state. Additionally, the material will be limited by the surrounding material's cooling shrinkage. At this point, the cladding layer will exhibit residual stress, and when this stress surpasses the material's strength limit, cracks will develop.[137,138] Corrosive media can enter the alloys through pores or cracks, making it is easy to produce local acidity in the pores or cracks because of the relatively confined environment, which accelerates alloy corrosion and leads to material failure.[137] Therefore, alloys or coatings used in corrosive media must be completely free of pores and cracks.[137]

On the other hand, the electrochemical behavior of NiTi was influenced by the grinding time, where the samples produced with a longer milling time demonstrated remarkable corrosion resistance due to the formation of a stable passive coating, which qualified them for use in bone implants.[25] Loukil and Fraj[66] explored how the NiTi SMA alloy corroded at different aging temperatures. According to OCP measurements, aged NiTi SMAs in 0.9% NaCl behave passively. Also, the presence of Ni_4Ti_3 precipitates and decreased particle size in the 450°C aged NiTi improves its corrosion characteristics compared to the as received (a j_{corr} drop of 87%). Furthermore, Guarise et al.[139] confirmed that the microstructural characteristics brought about by the AM process and heat treatment improved the mechanical and corrosion features of NiTi alloy.

Numerous surface modification methods, including magneto-electropolishing, laser irradiation, surface coating treatment, and vapor deposition, have been used to improve the biocompatibility and corrosion resistance of NiTi alloy implants.[140–144] Wang et al.[140] present in their research a new technique for creating a hydrophilic nanoporous functional surface using a combination of annealing and dealloying procedures to enhance the biocompatibility and corrosion resistance of NiTi materials. The findings indicate that while annealing encourages the densification and oxidation of the dealloyed layer, increasing the energy barrier for ion transit during corrosion, TiO_2 production during dealloying treatment significantly boosts material corrosion resistance. Furthermore, the sample's corrosion resistance improves as the dealloying temperature rises because of decreased surface imperfections.[140] Gill et al.[141] used the magneto-electropolishing approach to alter the NiTi alloy's surface, improving the material's cellular viability and resistance to corrosion. Witkowska et al.[142] enhanced material corrosion properties in a harsh environment by creating a hybrid a-CNH + TiO_2 + TiN-type surface layer on the NiTi surface using chemical vapor deposition and low-temperature oxynitriding. Yang et al.[143] modified the NiTi alloy's surface by combining polydimethylsiloxane modification with nanosecond laser irradiation to produce super-hydrophobic surfaces that enhanced the material's ability to withstand corrosion. Lin et al.[144] manufactured a coating on the surface of NiTi alloy that contained hydroxyapatite, which improved coating adherence and corrosion resistance in Hank's balanced salt solution.

The electrochemical reaction of NiTi alloys can be affected by changes in test media, such as PBS and Hank's solution, because of variations in ionic composition.[145] Compared to the other simulated bodily fluid solutions, Hank's solution exhibited the least aggression, as shown by the evolution of small current densities and the less active corrosion potential.[145] This behavior is linked to the presence of KH_2PO_4 and Na_2PO_4 in Hank's solution, which helps the biomaterials resist corrosion.[145] Notably, the concentration of Cl^- ions is lower in the Hank's solution compared to PBS, which is significant because higher chloride concentrations are known to encourage pitting corrosion in NiTi alloys by reducing the pitting potential.[145]

Apart from the research mentioned earlier, studies that examine NiTi's electrochemical behavior in simulated body fluids (SBFs) and at different pH values offer important insights into how well it performs in vivo in a range of biological circumstances. Huang et al.[146] reported that the release of Ti and Ni ions varied based on the specific manufacturer of the NiTi, as well as the acidity (pH level) and immersion duration in artificial saliva. Similarly, Rondelli et al.[147] explored commercial NiTi orthodontic wires in 0.9% NaCl and artificial saliva, demonstrating that corrosion resistance and pitting potentials are dependent on the alloy type and fluid composition.[147] Additionally, Pound's[148] investigation on NiTi revealed that reducing the pH with a constant chloride concentration (0.6% NaCl) thins the TiO_2 oxide layer from 5.3 nm to 4.2 nm, making

it more susceptible to pitting corrosion. Šušteršič et al.[149] focused on testing pH in SBF; they found that the corrosion varied by pH, where an increase in pH leads to an increase in corrosion. Jia et al.[150] reported that NiTi alloy fabricated by selective laser melting shows excellent corrosion resistance in NaCl physiological solution but noted that the presence of fluoride ions (F^-) and low pH in simulated oral environments can significantly affect corrosion behavior. The increase in fluoride concentration and the decrease in pH value could accelerate the corrosion of the SLM sample.

In summary, based on the studies discussed previously, the NiTi alloys exhibit good corrosion resistance in simulated body fluid due to the stable and protective oxide layer that forms, which effectively limits ion interaction with the environment. The stability of NiTi alloys' passivation layer is influenced by several important factors, including grain size, the oxide layer's thickness and composition, and surface treatments such as thermal oxidation, chemical etching, and electropolishing.[151,152] Among these, the electropolishing process was found to be beneficial because it removes the outermost layer of the alloy's surface that forms during metalworking operations and successfully lowers surface roughness from the micro- to nanometer level, enabling the surface to function as a biomaterial.[153] According to Venugopalan et al.,[154] electropolishing NiTi produces mostly titanium oxide, which is a key component in enhancing corrosion resistance by homogenizing the surface layer's thickness, topography, and chemical composition.

Bae et al.[155] proposed that heat treatment is better carried out in a salt furnace rather than an air furnace, and passivation can be a helpful way to boost corrosion resistance by suppressing the oxidation process. Also, as the grains are finer, more grain boundaries, which serve as passive film nucleation sites, encourage the creation of a more compact and noble passive film.[25] On the other hand, research indicates that the surface oxide layer's thickness and composition have a significant impact on nickel ion release in NiTi medical devices.[156,157] Furthermore, the passivation of NiTi alloy is morphologically composed of two layers: the inner layer is compact, whereas the outer one is porous .[25,158] The outer porous layer promotes osteointegration, while the inner layer, which is thicker and prevents metal breakdown, is primarily responsible for the corrosion resistance.[25,158]

MICROSTRUCTURAL AND MECHANICAL PROPERTIES OF NITINOL ALLOYS

The microstructural and surface properties of biomedical NiTi alloys have a significant influence on their mechanical and tribological properties, including ductility, strength, hardness, and wear resistance. There is a direct correlation between microstructural evolution and mechanical performance. In particular, hardness is an important factor in influencing the friction and wear properties of NiTi-based materials.[19] Processing techniques can significantly alter the microstructure and, consequently, the mechanical properties. For instance, Zhou et al.[159] report that the microhardness of NiTi alloys increased from 118 HV to 206 HV by refining the grain and hardening the work using the constrained groove pressing (CGP) process. Additionally, improving the grain and dislocation boundaries contributes to improved microhardness. The flow stress and microhardness value increase when twin regions and dislocation structures form, preventing slip motions.

These enhancements will reduce the need for additional supporting elements to increase the NiTi alloy's load-bearing capability and structural integrity in biomedical applications.[159] Similarly, Bouchareb et al.[26] revealed that the grain refinement of NiTi alloys obtained from the mechanical alloying increased hardness from 201 HV to 341 HV because of frequent cold welding, severe plastic deformation, and fracturing of powder particles during the mechanical alloying process, all of which enhanced their resistance to wear. In another recent study, Chen et al.[160] investigated the impact of ultrasonic surface shot peening (USSP) on the microstructure, hardness, and tribological behaviors of 50.8Ni-Ti SMA alloy. Their results show that by improving the microstructure and encouraging the martensitic transition in the surface layer, the USSP treatment increases surface hardness. A maximum hardness of 541 HV was obtained with a test load of 25 g. Remarkably, USSP treatment can improve wear resistance and reduce the coefficient of friction (COF) at sliding wear speeds as low as 5 mm/s. Furthermore, Yin et al.[161] found that phase transformation is suppressed and hardness takes over as the primary factor governing wear resistance when grain size falls, improving tribological performance.

The microstructure of the NiTi alloy changes from B2 austenite, the parent phase of martensite, to B19' martensite, the monoclinic martensitic phase, under external stress and temperature changes. This alters the material's superelasticity and shape memory, allowing it to change shape for a range of applications.[27] In addition, temperature and strain rate are two external factors that can affect the microstructure of NiTi SMAs. Changes in strain rate during loading and unloading cycles can affect the heat produced during phase change, which affects the material's mechanical and functional characteristics.[162]

On the other hand, the addition of alloying elements has a major effect on the mechanical properties, especially hardness and microstructure. Yuan and colleagues[163] examined the effects of vanadium (V) addition on the microstructure, mechanical properties, and oxidation performance of 60NiTi. The microstructure and properties of very

Table II. Hardness of the NiTi alloys under different processing conditions

Condition/treatment	Hardness (HV)	Ref.
Mechanical alloying at different milling times 2, 6, 12, and 18 h + sintering	201 to 341	26
Ultrasonic surface shot peening	Up to 541	160
NiTi-Hf alloys (aging + precipitation)	769 to 707	164
Post-weld heat treatment (PWHT) at 650°C and 850°C	375 to 493	165
Spark plasma sintering then annealed at 600°C	504 to 509	166
Constrained groove pressed	Up to 206	159
High-pressure torsion (HPT)	456.8 ± 14.9	167

Ni-rich NiTi alloys were found to be improved by a V content of < 3 at.%, and the transition from the Ni_4Ti_3 phase to the Ni_3Ti_2 and Ni_3Ti phases was enhanced by the additional V. The authors claim that the alloy's hardness increased as its V content grew because of the solid solution strengthening action of V and the strengthening effect of the Ni_3Ti_2 phase, whereas its tensile strength and fracture toughness first climbed before declining. Likewise, Zhou et al.[27] report that the addition of tantalum (Ta) significantly refines the grains of the NiTi alloy, causing a 54.5% decrease in columnar crystal spacing and a 44.1% decrease in equiaxed grain diameter. Precipitation hardening and precipitation phase strengthening will also significantly increase the alloy's strength and hardness. Table II summarizes the hardness of NiTi alloys under different processing conditions.

TRIBOLOGICAL BEHAVIOR OF NITINOL

Many biomedical applications depend on tribological characteristics, which are often linked to wear, friction, and lubrication.[9,168–173] Although the implant's mechanical characteristics dominate the bone-implant interface's structural behavior, it is crucial to comprehend the frictional behavior between the implant and bone since it ultimately helps ensure the implant's primary stability.[169,174] Tribology has several uses in the biomedical industry, including the wear of the femoral stem and acetabular cup in hip prostheses, plates and screws in bone fixation, implants for dentures, and heart valves.[169,174] For these reasons, the materials ought to be resistant to wear and have a low coefficient of friction to endure the continuous abrasion of implants against bodily tissues.[169]

Mechanical characteristics, including toughness, hardness, work hardening, strength, and many others, determine a material's ability to withstand wear.[20,25,175–178] Nevertheless, it seems that these features are not the leading cause of the outstanding wear resistance of NiTi alloys. The recovery of the superelastic deformation is thought to be the primary causes of shape memory NiTi's remarkable wear resistance,[179–184] although it depends on the temperature and alloy composition.[185]

However, some studies contradict this opinion. For instance, Ball[186] ascribed Ti-Ni's wear characteristics only to work-hardening. Additionally, Hattori and Tainaka[187] showed that the martensite phase has a higher resistance to erosion than the austenite phase. According to Liang et al.,[188] relating NiTi alloy wear behavior exclusively to pseudoelasticity is impractical. Their findings indicate that the recoverable strain limits the total of the pseudoplastic and pseudoelastic strain limits, which is the primary determinant of NiTi alloys' resistance to wear. Further study is necessary to understand the tribological characteristics of the austenite and martensite phases of NiTi alloys.[189] In tribological testing, tribometers are employed to assess tribological properties such as the coefficient of friction and wear rate. Although many various kinds of tribometers are available today, we have considered the most widely used ones, including block-on-ring, four balls, pin-on-disc, and ball-on-disc, etc. (refer Fig. 8).

Tribological tests have basically been carried out in accordance with ASTM G133, ASTM G99, and ASTM F732 testing standards.[190,191] In compliance with the ISO-14242 and ISO-14243 series, the implants are tested using cross-shear motion at a speed of 1 Hz or less using serum; ISO 14243 outlines the relative angular movement between articulating components, the pattern of applied force, the testing speed and duration, sample configuration, and test environment to be used in wear-testing machines with load control for total knee-joint prostheses.[190]

According to recent research, NiTi alloy exhibits superior wear resistance compared to other traditional materials such as Ni-based, steels, and Co-based tribo-alloys.[26,178,192–194] Li and Liu[195] conducted sliding wear experiments on 2Cr13 steel and Ti-50.3 at.% Ni alloy. They discovered that the NiTi alloy is far more wear-resistant than steel. Furthermore, Farhat and Zhang[192] utilized a pin-on-disc setup to study the sliding wear behavior of superelastic NiTi alloy on superelastic NiTi, pure Ti, and Ni. Compared to pure Ti and Ni, the superelastic NiTi alloy exhibited roughly 30 and 10 times higher wear resistance. Also, superelastic

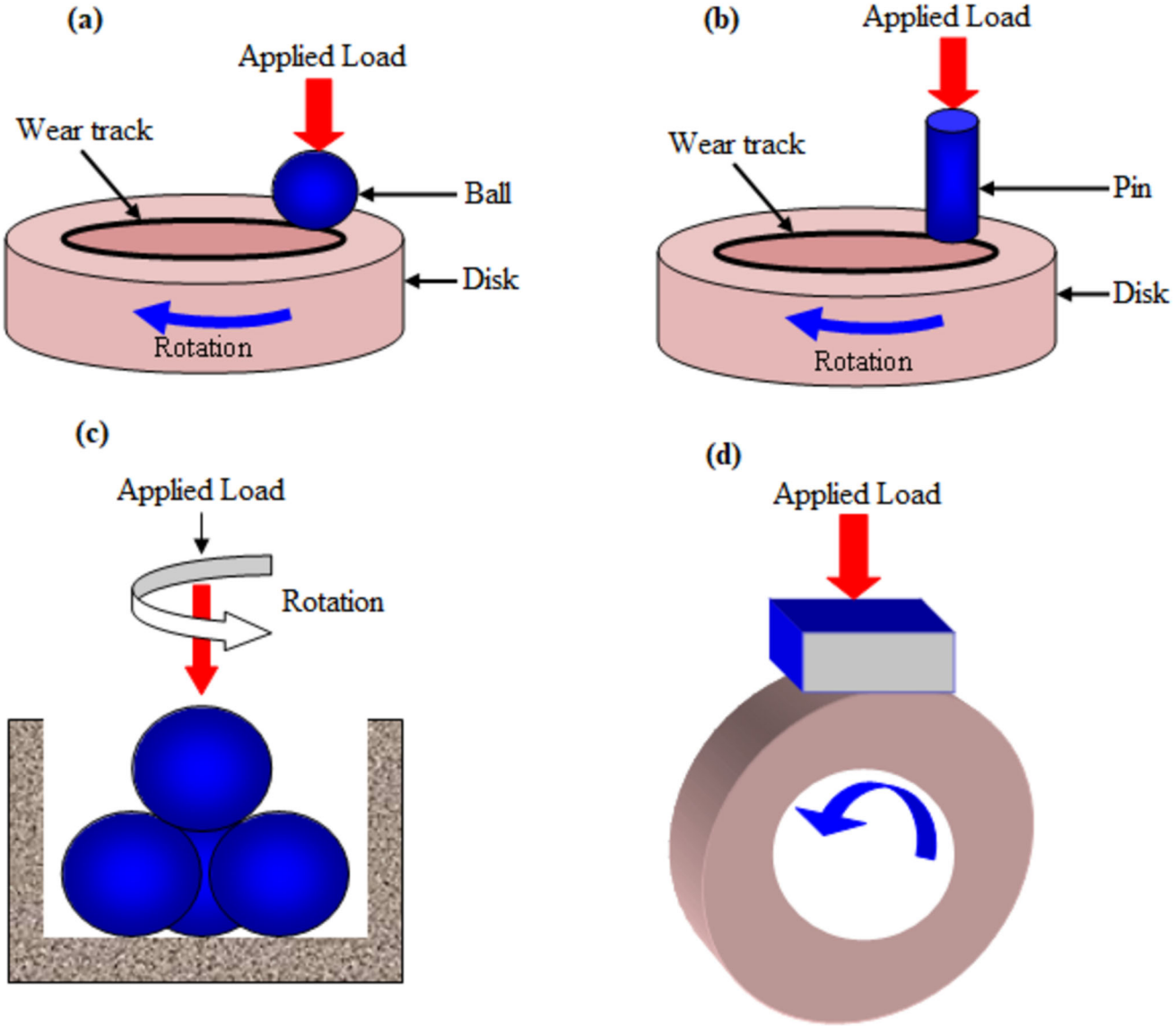

Fig. 8. The common tribometers used in the tribological tests: (a) ball on disc; (b) pin on disc; (c) four balls; (d) block on ring.

NiTi revealed better resistance to wear than AISI 304 stainless steel, although it has comparable hardness.[196]

Jin and Wang[197] studied the wear resistance of several alloys and found that NiTi SMAs are more resistant to wear than Co45 and 38CoCrMoAl alloys at high hardenability, especially under high loads. Huang et al.[198] investigated the wear performance of NiTi alloys obtained by laser powder bed fusion additive manufacturing. The influence of sliding speeds and heat treatment on wear resistance was investigated. Heat-treated samples were shown to be harder, while as-built samples showed greater toughness and transformation strain, resulting in several wear modes that depend on frictional heat release, including adhesive, abrasive, and delamination wear. Furthermore, the wear behavior of NiTi is greatly influenced by the normal load and sliding distance during the test; as documented by Abedini et al.,[199] the wear rate reduced as the normal load and sliding distance rose. The material's resistance to wear was increased because the high plastic yield stresses of the low-temperature phase of converted martensite in NiTi inhibited plastic deformation. Also, the combination of a lower Young's modulus, considerable recoverable transformation strain, low transformation stress, and high plastic yield strength of martensite contributes to this enhanced wear resistance, as highlighted by Chrzanowski et al.[200] The improved tribological features of NiTi alloy (low COF and excellent wear resistance) are related to their high density and hardness, as reported by Bouchareb et al.[26] The wear rate and COF of several titanium alloys under various test conditions with the NiTi alloys produced in our previous study[26] are displayed in Table III.

Table III. Wear rate and friction coefficient (COF) of various types of titanium alloys

Materials	Testing conditions	Hardness	Wear rate	COF	Ref.
CP-Ti	Scratch tests Load of 1 N and 5 N under dry sliding conditions and room temperature, with a 5 mm scratch length and a 2.5 mm s^{-1} tangential scratch speed	146 $HV_{0.5}$	Not specified	For 1 N load = 0.63 ± 0.11 (avg.) For 5 N load = 0.64 ± 0.06 (avg.)	201
$Ni_{50}Ti_{50}$	Tribometer: ball-on-plate Normal loads of 2, 10, and 20 N with a 5 mm stroke length and a 25 mm s^{-1} sliding speed	201-341 HV	29.67×10^{-3} μm^3/N μm (avg.)	0.31 (avg.)	26
Ti-6Al-4V	Tribometer: pin-on-disk Normal load: 5N Sliding speed was changed from 0.2 to 0.8 m s^{-1}	350 ± 5 $HV_{0.5}$	Not specified	For 1 N load = 0.65 ± 0.09 (avg.) For 5 N load = 0.46 ± 0.04 (avg.)	201
Titanium alloy grade 23	Tribometer: pin-on-disk Normal load ranging from 30 to 50 N, with a sliding distance between 1000 and 2000 m, and a sliding speed of 1 to 2 m s^{-1}	Not specified	Not specified	0.3308 (avg.)	202
Ti-17Nb-6Ta	Tribometer: pin-on-disk Loads of 10, 15, and 20 N were applied, with a 10 mm track length and 2 Hz frequency	176 HV	0.016 mm^3/N m (avg.)	0.22 (avg.)	203
Ti-25Nb-xZr	Tribometer: ball-on-plate Varied loads of 2, 10, and 20 N at a constant speed of 15 mm s^{-1}	233-300 $HV_{0.02}$	For 2 N load and Zr content 30 at.% = 6.23×10^3 μm^3/N μm (avg.)	For 2 N load and Zr content 30 at.%: ranged from 0.3 to 0.5 (avg.)	9
$Ti_{13}Nb_{13}Zr$	Tribometer: ball-on-plate The normal load of 29.4 N, the frequency was set at 1 Hz, while the stroke length was fixed at 6 mm	≈ 225 HV	1.36×10^{-4} mm^3/N m (avg.)	≈ 0.15	204
Ti-20Mo	Tribometer: ball-on-disk Applied loads of 2, 8, and 16 N were used at a sliding velocity of 10 mm s^{-1}	255 $HV_{0.02}$	For 2 N load ≈ 4.21×10^3 μm^3/N μm (avg.)	For 2 N load ≈ 0.20 (avg.)	7
Ti-3Fe	Tribometer: ball-on-disk The normal load of 2 N with velocity of 0.08 m/s, and a sliding distance of 300 m	Not specified	5.55×10^6 mm^3/N m (avg.)	0.55 (avg.)	205
Ti-4.5Al-4Fe	Dry sliding experimental technique using the tribometer The normal loads of 10 N and 20 N with a linear speed of 0.06 cm/s	Not specified	For 10 N load = 1.033×10^{-5} mm^3/N.m For 20 N load = 1.74×10^{-5} mm^3/N.m	For 10 N load = 0.047 For 20 N load = 0.015	206
Ti35Zr28Nb	Using a UMT-II multi-functional friction and wear tester Load of 3 N with slip speed of 0.03 m/s (slip frequency was 1 Hz) and the stroke length of 15 mm	310 ± 1.8 HV	0.0021 ± 0.0002 mm^3/N m (avg.)	0.42 (avg.)	207
Ti matrix composites reinforced with TiB_2	Tribometer: ball-on-disk The normal load of 9 N and 13 N with the linear speed of 5.17 cm/s and 11.41 cm/s.	Range: 228.5 HV_1 to 304.8 HV_1	For 9 N load ≈ 0.0008-0.00108 mm^3/N.m For 13 N load ≈ 0.0008-0.0014 mm^3/N.m (avg.)	0.40-0.54 at 9 N 0.65-0.75 at 13 N (avg.)	208

Table III. continued

Materials	Testing conditions	Hardness	Wear rate	COF	Ref.
Ti-6.5Al-3.5Mo-1.5Zr-0.3Si	Dry sliding wear tests, Tribometer: pin-on-disk Normal load of 50, 100, 150, 200, and 250 N, the sliding velocity of 1 m/s	360 HV under the load of 250 N at 673 K	For 250 N load and temperature of 673 K (400°C) $\approx 200 \times 10^{-6}$ mm^3/mm (avg.)	Not specified	209
Ti20-Al16-V16-Fe16-Ni16-Cr16 high entropy alloys	Tribometer Normal load of 10 and 20 N with duration of 900 s, a frequency of 15 Hz, and a linear speed of 0.06 cm/s Tested on dry and wet modes	Not specified	8.2×10^{-3} to 1.3×10^{-4} mm^3/N m (avg.) depending on sintering temperature	0.036 (avg.) at 1000°C; and 0.51 (avg.) at 700°C	210

Compared to other alloys, including CP-Ti, Ti-6Al-4V, Ti-17Nb-6Ta, and titanium alloy grade 23, it is evident that the NiTi alloy has the lowest COF (0.31).[26] This drop in COF suggests that NiTi has a high wear resistance level, making it more resilient for applications such as implants. Furthermore, the alloy's low coefficient of friction raises the possibility that it will reduce friction during operation, reducing surface wear and improving material performance in human environments. These results suggest that NiTi is more effective in medical applications regarding reduced friction and wear resistance, extending the material's life and reducing the need for frequent replacement or revision surgery.

Material wear is undesirable in biomedical applications because it affects the longevity and performance of medical devices like implants and prostheses.[211] In addition, wear contributes to wear debris and is a primary cause of component failure. Wear debris is generally unwanted in any tribological system since it pollutes the environment and has a range of effects on the wear process (third body effect).[211] The primary cause of patients' long-term problems in biomedical applications, including total joint replacement, is the created wear debris.[211] Wear debris can cause inflammatory responses when discharged into the surrounding tissue, leading to bone resorption and possibly implant loosening.[212–215] Consequently, enhancing the tribological characteristics allows implants to last longer and, as a result, reduces the need for repeated surgery.[213,215–217] The deposition of coatings or microstructural alterations that alter the material's characteristics can improve wear resistance.[213,215] Figure 9 shows the wear debris produced because the alloy surfaces' oxidized and work-hardened layers deteriorate and surface asperities are destroyed. These processes invariably result in a third body contributing to surface deterioration, fostering the wear mechanism, including delamination, adhesion, and abrasion.[26]

BIOCOMPATIBILITY OF NITINOL

Biocompatibility Issues of Nitinol

Biocompatibility is a key property of biomaterials that makes them appropriate for interacting with biological systems while avoiding adverse effects. The European Society for Biomaterials stated that the accepted definition of biocompatibility is "the ability of a material to perform with an appropriate host response in a specific application."[218,219] To complete this explanation, the following three conditions are necessary:[219]

- A material must perform its necessary functions and not merely be present in the tissue;
- The induced reaction needs to be suitable for the intended use;
- The basis for responding to that specific material and its suitability must differ from others.

Kohane and Langer explicated biocompatibility concerning drug delivery and described biocompatibility as an "expression of the benignity of the relation between a material and its biological environment."[220] In biomedical applications, the biocompatibility of materials has been restricted to the absence of corrosion or any degradation process.[221,222] Figure 10 depicts factors affecting biocompatibility.

In medical practice, two categories of biocompatibility exist structural biocompatibility and surface biocompatibility.

- **Structural biocompatibility**: The interplay between the biomaterial's structure and the biological system's characteristics. According to these interactions, a biomaterial has a certain capacity to carry out a specific task inside a medical device.[223]
- **Surface biocompatibility**: Describes the implant's compatibility (topographically and chemically) with the cells around it.[223]

Generally, biocompatibility refers to materials that are introduced into the human body without

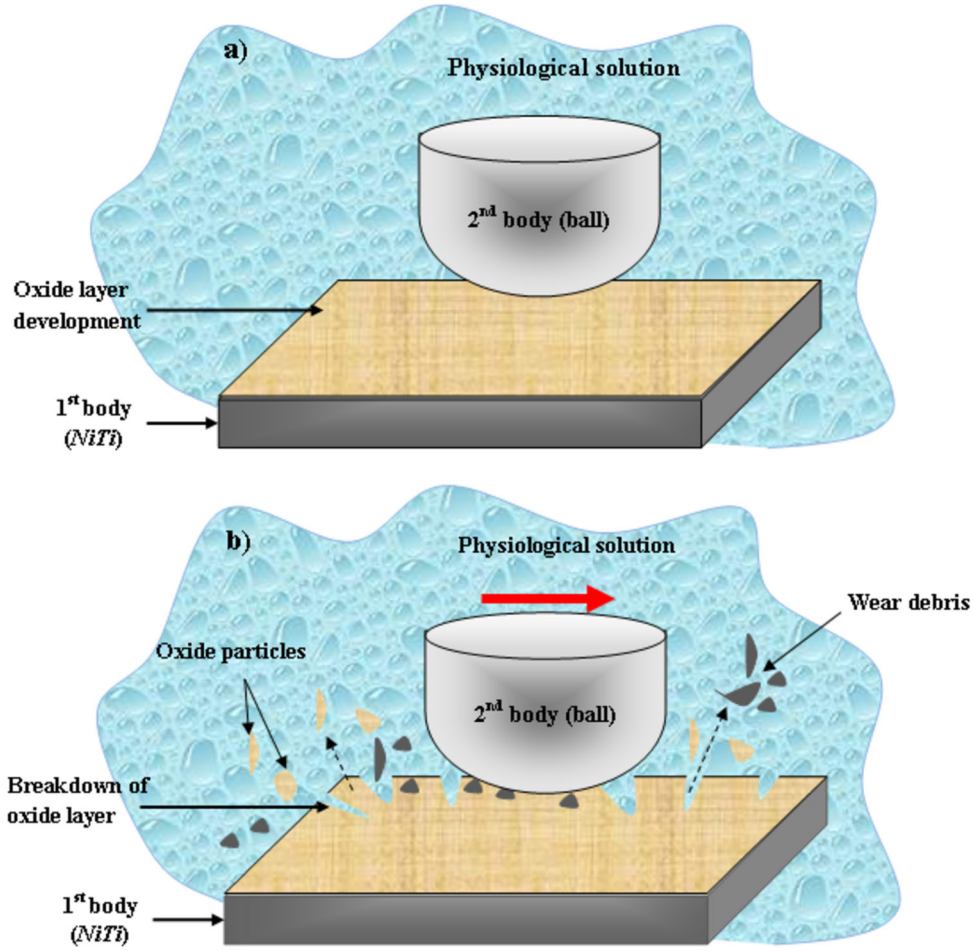

Fig. 9. Schematic showing the tribological interaction results between NiTi alloy and a ball surface: (a) before the sliding; (b) after the sliding.

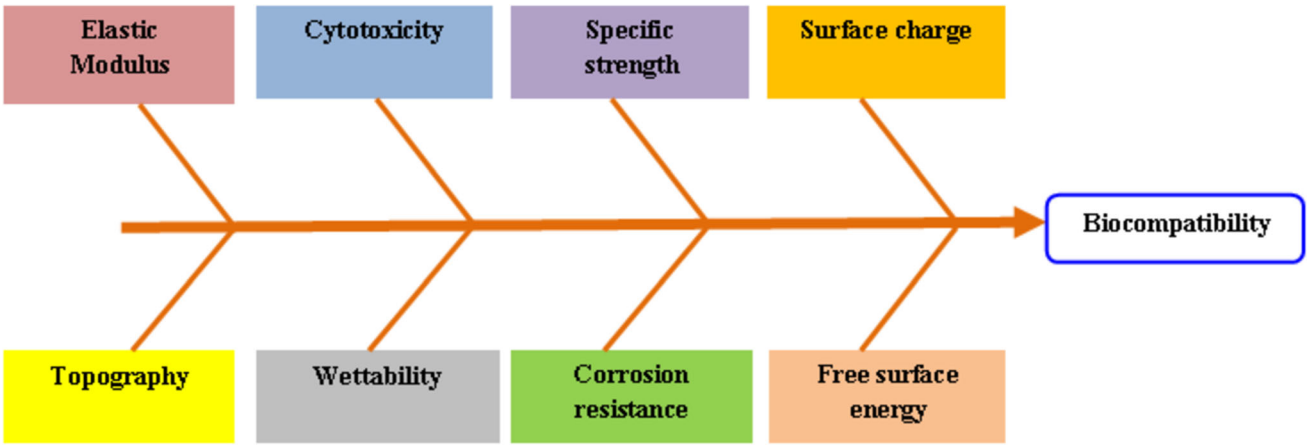

Fig. 10. Schematic presentation of factors affecting biocompatibility.

causing any adverse reactions.[224] The compound NiTi comprises roughly equal quantities of Ni and Ti. Ni release is a major problem with NiTi biocompatibility, demonstrating unfavorable biological effects, while Ti showed favorable biocompatible effects.[225,226] NiTi should be corrosion-resistant for optimal biocompatibility, meaning the least amount of nickel should be released.[50] Metal alloys' ability

to withstand corrosion depends on the passivation phenomena that result from producing a metal oxide layer on the metal's surface.[25,32]

The kinetics of nickel ion release from NiTi alloys are influenced by a wide range of factors, including the manufacturer, surface condition, immersion duration, and pH of the physiological environment.[24,227,228] According to several studies,[145,229,230] the release rate is higher during the initial exposure period and gradually decreases as a stable oxide layer forms on the surface.

Nickel ions are a biological concern because they oppose vital metal ions, including calcium, magnesium, and zinc ions, necessary for several biological activities like cell reproduction and protein synthesis.[23] In addition, excessive nickel ion release has been linked to allergic reactions, malignancies, and inflammatory responses.[231] However, their release into physiological fluids is restricted by the strong interatomic bonding in NiTi alloys and the protective TiO_2 oxide layer on their surface.[23,25] This phenomenon explains why numerous investigations have verified NiTi's biocompatibility. To further enhance biocompatibility and minimize ion release, treatments such as oxidation, electropolishing, and coatings are commonly applied.[232]

Nevertheless, several experimental researchers investigated NiTi's biocompatibility and demonstrated that it is safe to use various biological sectors.[23,224] This results from developing a titanium-oxide layer over a chemically inert NiTi alloy.[224] This oxide layer increases the material's durability by preventing corrosion in the underlying layers.[224] The oxide layer also protects the material from nickel chemical and physical oxidation by changing the pathways involved in nickel oxidation.[224]

Studies conducted in vitro and in vivo showed that passivated NiTi was biocompatible and did not cause genotoxic or toxic effects when it came into contact with a physiological environment, as Gill et al.[141] highlighted. Dahiya et al.[233] identified the production of passivated NiTi nanoparticles (NPs) by the laser ablation of NiTi targets, followed by various surface treatments. This work examines this process, highlighting that surface modifications improved biocompatibility and decreased nickel leaching. A review published by Mani et al.[234] discusses several surface finishing techniques, including mechanical polishing and electropolishing, and assesses how they affect the performance and biocompatibility of NiTi implants. Ryhänen et al.[235] investigated the cytotoxicity and NiTi corrosion rate in human osteoblasts and fibroblasts. Their findings demonstrated that NiTi exhibits favorable in vitro biocompatibility with both cell types. Furthermore, NiTi did not negatively affect cell survival, even though the rate of nickel ion release was originally higher. It did not stop cells from growing or hinder the growth of cells that came into direct contact with its surface.[235]

A study by Balakrishnan et al.[236] compared the biocompatibility of stainless steel and NiTi as bladder implant materials; the SEM investigation suggested that NiTi was more inert than stainless steel, and there was less fibrosis or inflammation. Castleman et al.[237] examined NiTi's toxicity to soft tissues by implanting it into canine muscular tissue. Clinical, radiological, and morphological assessments conducted 17 months post-implantation revealed no negative tissue reactions, suggesting that NiTi does not substantially damage soft tissue. In the Ryhanen investigation,[32] rats' paraspinal muscles and nearby sciatic nerves were implanted with NiTi to evaluate their soft tissue toxicity with two other biomaterials (Ti-6Al-4V, stainless steel). The general inflammatory reaction was similar to the other two alloys, and the examination revealed no indications of granulomas or necrosis in the muscle tissue. Additionally, there was no discernible tissue reaction from the sciatic or peripheral nerves.[32]

The low-temperature dealloying treatment method improves the biocompatibility of NiTi shape-memory alloys while preventing sensitization and allergies, making it a potential technique for biomedical applications. This method set the stage for clinical trials on NiTi surface modification, as reported by Huang et al.[238] The outcomes attained by Kujala et al.[239] indicate that NiTi suture is a promising novel material for tendon sutures, having superior strength qualities and good biocompatibility in tendon tissue compared to the traditional materials used in tendon surgery.

On the other hand, diamond-like carbon (DLC) film is well known for its chemical resistance and biocompatibility.[50] DLC film coatings increase the durability of biomaterials and shield them against deterioration.[50] Yeung et al.[240] analyzed the impact of nitrogen plasma implantation on NiTi alloys. They discovered that the TiN layer formed during this method had outstanding resistance to wear and corrosion. Furthermore, the TiN layer inhibits nickel ion release.[240] Compared to pure titanium, coatings of titanium-silver alloys have greater corrosion resistance.[50,241]

Biocompatibility Tests

To guarantee the prototype's safety for subject usage, biocompatibility testing must be carried out in addition to mechanical and functionality testing.[242] This testing ensures that when the prototype comes into contact with mucosal tissue, it will not trigger an undesirable immune response that causes inflammation, discomfort, and infection.[242] Below are some of the most widely used biocompatibility test techniques, including cytotoxicity, sensitization, irritation, genotoxicity, hemocompatibility, and thrombogenicity tests.

Hemocompatibility Test

A hemocompatibility test can identify and assess the thrombus reaction brought on by a foreign body or foreign substance (in our case, NiTi) that comes into contact with blood.[232,243] A foreign substance inserted in the body causes platelet activation and blood coagulation.[232,243] Therefore, early in the biomaterial's development, in vitro investigations, such as hemocompatibility tests, aid in identifying and eliminating undesirable foreign substances and materials that cause clotting.[232] Thus, heparin and sodium citrate are commonly utilized as anticoagulants to assess Ti-Ni-based biomaterial's hemocompatibility.[232]

Cytotoxicity Testing

In a cell culture assay, cytotoxicity tests are regarded as screening assays that evaluate the response of the living cell to the implant, including cell survival and growth potential.[244] ISO 10993-5 describes the in vitro cytotoxicity tests.[245] The standard outlines three distinct methods: direct, extract, and indirect contact tests.[245] Cytotoxicity testing aims to ascertain the biological response of mammalian cell lines after coming into contact with the substance or its extract.[246] The test begins with the growth of cells on plates.[246] The cells are either produced from mouse fibroblasts or Chinese hamster ovaries, depending on the test being conducted. The test article or test article extract is introduced to the cell population once the cells have reached the proper confluence or density. There is a time after exposure when the plates are checked for signs of cell death.[246]

Cells are extremely sensitive to even slight pH, salinity, or temperature changes, which will kill them. A certain degree of this kind of cell death can happen without the test item being classified as cytotoxic. The test item will be classified as cytotoxic if most of the cells are either dead or dying. In vivo and in vitro tests must be performed on a testing battery.[246]

The NiTi alloys are considered non-cytotoxic according to ISO 10993-5 if the cell viability remains > 70%. According to Ibrahim et al.'s study,[226] both MG63 and H9C2 cells demonstrated non-cytotoxic behavior by maintaining cell viability levels > 70% on days 1, 3, and 5. Furthermore, Chmielewska et al.[247] confirmed the remarkable biocompatibility of the alloy by showing that NiTi samples had 99-102% survivability at different dilution levels.

Sensitization and Irritation Test

These tests are designed to detect allergic reactions involving the immune system without the involvement of an immunological mechanism, as well as the localized inflammatory response to a single, repeated, or continuous exposure to the test item.[248] These tests are advised by ISO 10993 Part 10.[248]

Sensitization: assesses the possibility of contact sensitization since even minute quantities of material can cause sensitization or allergic reactions when they come into contact with skin and other tissues.[249] Sensitization tests must be performed on each permutation.[249]

Irritation: uses locations such as the skin, eyes, and mucous membranes to assess the likelihood of irritation in an appropriate model.[249] Every other permutation must be examined for irritation, except for long-term indirect ECDs.[249]

Genotoxicity

Genotoxicity determines whether a substance causes gene mutations, alterations in the number or structure of chromosomes, or other DNA or gene toxicities due to exposure to the test material using cell cultures or other methods.[249] Chromium is one genotoxic substance that can interact with DNA and cause cancer.[249]

Lysis and Thrombogenicity Test

Thrombogenesis is the propensity for a clot to form when an implanted NiTi comes into contact with blood.[232] Laboratory testing and research have been conducted to assess acute coronary syndrome's thrombogenicity to comprehend the physiology of the condition.[232] The ability of devices and materials to generate thrombus markers during static and dynamic flow is assessed by thrombogenicity in vitro testing, which is a component of regulatory approval. These tests are frequently conducted to evaluate hemodialyzers and blood-recirculating systems.[232] A summary of NiTi's biocompatibility investigations employing various testing methods is provided in Table IV.

MEDICAL APPLICATIONS OF NITINOL ALLOYS

NiTi shape memory alloy is an important biomaterial owing to its unique features, such as superelasticity and shape memory effect, excellent resistance to corrosion, and biocompatibility, making it widely utilized in the biomedical area. Table V summarizes the use of NiTi alloys in medical procedures.

Applications of Nitinol as Arch Wire

For more than 2 decades, dentists have utilized NiTi as an orthodontic arch wire.[50,256] In orthodontics, NiTi can be used because of its unique mechanical properties, such as superelasticity (pseudoelasticity) and shape memory effect.[257,258] When utilized as an orthodontic arch wire in saliva solution, this alloy resists pitting better than stainless steel.[50]

Table IV. Summary of biocompatibility studies on NiTi alloys using different testing techniques

Testing technique	Key findings	Quantitative results	Cell line/assay	Ref.
Cytotoxicity and corrosion	In vitro, NiTi is biocompatible with human fibroblasts and osteoblasts	The proliferation of fibroblasts (FB): 108%. The proliferation of human osteoblasts (OB): 101%. Initial Ni release: 129–87 μg/L	Human fibroblasts (FB), osteoblasts (OB); cell proliferation assay	235
Cytotoxicity and adhesion assays (ISO 10993-5)	High viability; XY-NiTi surface promotes better adhesion	Viability > 90%;	Bone marrow mesenchymal cells (BMMCs)	250
Cytotoxicity	The biocompatibility of NiTi is superior to that of ternary alloys	NiTi42Cu7: Cu release ≈ 2.8 μg/cm^2 in 7 days; higher cytotoxicity observed	Cultured epithelial cells from human gingiva	251
Cytotoxicity of corrosion products	Corrosion products from NiTi stent wires inhibited smooth muscle cell growth by disrupting the G0-G1 phase transition of the cell cycle, highlighting the importance of surface treatments to reduce toxicity	Nickel ion release around 0.95 ± 0.23 ppm has been linked to mild cellular stimulation, while concentrations exceeding 9 ppm have been shown to significantly inhibit cell growth	Rat aortic smooth muscle cells; [(3) H]-thymidine uptake, cell cycle analysis	252
Cytotoxicity (ISO 10993-12)	Cytotoxicity studies demonstrated the investigated NiTi alloy's safety and biocompatibility in its as-cast state	Cell viability > 70%	MG63 and H9C2	226
Cytotoxicity (ISO 10993-5)	Both materials non-cytotoxic	Cell viability ≥ 95.5% at all dilutions	L929 fibroblasts	247
Cytotoxicity	Surface modification using hydrogen peroxide, anodization, and acid etching was found to be effective in waning the nickel leaching and improving biocompatibility	Tolerated up to 100 ng/μL when the percentage of viable cells were expressed considering untreated cells as 100% viable	MG63 osteoblasts; MTT assay and ICP-MS	233
Cytotoxicity (ISO 10993)	$NiTi_{50.8\%}$ single crystals do not trigger a cytotoxic reaction	Nickel ions < 1 ppm Cell viability ≥ 100%	L-929 mouse fibroblasts, J-774 macrophages; MTT assay	253
Cytotoxicity (ISO 10993-4)	Annealing at 500–700°C worsens cytocompatibility compared to the control sample, but annealing at 800–1000°C improves cytocompatibility	Cytotoxicity NiTi500 = 61 ± 7%, NiTi600 = 74 ± 7%, NiTi700 = 58 ± 6%, NiTi800 = 14 ± 7%, NiTi900 = 21 ± 4%, and NiTi1000 = 28 ± 4%	MCF-7 cells; MTT, microscopy	254
Genotoxicity/cytotoxicity (10993-5)	Compared to NC, NiTi has superior biocompatibility	-	Fibroblasts; MTT assay	255

Superelasticity allows NiTi arch wires to revert to their pre-loading shape upon unloading.[259] The alloy has a 7-8% strain twisting capacity, more than 40 times that of stainless-steel wire.[259] While loading causes the periodontal ligament to contract and the NiTi alloy's crystalline structure to return to its initial shape, mobilization begins at a plateau of light forces known to provide the best biological tooth movement.[259,260] NiTi alloy can "remember" its original shape after being bent elastically or pseudo-plastically by raising its temperature because it is a material with a shape memory effect. A body-centered cubic phase (austenitic phase) changes into an orthorhombic or monoclinic martensite phase because of thermoelastic martensitic transformation.[259] As a result, the distorted orthodontic arch wire moves and aligns the teeth by taking on their original shape in orthodontics. Throughout the spectrum of arch wire activation, the stress stays nearly constant, delivering a minimal level of force that results in a good reaction from the surrounding tissues, increased patient

Table VI. Applications of NiTi alloys in medical devices

Field	Medical device	Applications	Ref.
Orthopedic	-Total hip replacement - Orthopedic instrument	-Hip joint reconstruction/arthroplasty - Pressurized riding nails, polypatellar devices, frame-type internal fixators	25, 26, 224, 264
Vascular/venous	Stents	Treatment of chronic venous obstruction and femoropopliteal arterial disease	265, 266
Dental	Stomatological instruments	Orthodontic devices: braces and wires	264, 267, 268
Surgical instrument	- Nitinol-actuated laparoscopic tool - Micro-forceps	- Laparoscopic tool for renal denervation for the treatment of resistant hypertension - Micro-forceps for retrieval of intravascular objects	269, 270
Minimally invasive therapeutic device	Guide wires	Used for cardiovascular magnetic resonance catheterization	271

comfort, and more physiological tooth movement.[259] Root resorption can be caused by high orthodontic forces, which is unwanted since it may compromise the tooth's long-term health.[259]

Applications of Nitinol in Stents

These alloys can be employed as stents and blood clot filters in cardiovascular therapies since they undergo phase transformation and alter structure.[50,261] Peripheral artery stents are subjected to substantial mechanical stress from the environment, such as when walking, running, or bending the knee.[262] NiTi is more resilient to these external forces than other materials because of its special properties of superelasticity and stress hysteresis.[262] These stents' kink resistance makes them perfect for the complex arterial courses of peripheral arteries.[262] Depending on how they are inserted, stents can be divided into two categories: self-expanding and balloon-expandable.[262,263] The diameter of a self-expanding stent is made greater than the diameter of the vessel. After that, a delivery catheter is crimped into it.[262,263]

Applications of Nitinol as Guided Wire and in Endoscopes

A guided wire is a tiny wire made of metal. It is inserted through a minor cut or a natural hole. The risk of harm decreases when using a NiTi-shape memory alloy in the guided wire. Endoscopes are among the most complex medical devices that use shape memory alloys.[50] The introduction of NiTi shape memory alloy significantly improved the flexibility and control of endoscopes.[50]

Applications of Nitinol in Orthopedics

NiTi is frequently used to internally fix bone in the hands, shoulders, spine, limbs, and other areas, maintaining the anatomical decrease in fractures.[224] The fracture type and site determine the depth and fixation location of different internal fixators.[224] For the internal fixator to play a fixed role during the bone tissue regeneration of the fracture site and reduce the stress concentration of the implant site, it must have the proper strength, exceptional elasticity, and deformation in the body.[224] NiTi can also be utilized as an implant since its elastic modulus isalso similar to human bone.[25,26,50] Additionally, NiTi has the following benefits:[224]

- Following the fracture end's fixation, micromotion will occur, which is consistent with the biomechanical properties of stress osteogenesis in humans.
- Short operating time, easy fixing technique, and decreased infection risk.
- The periosteum sustains less damage when the fracture end is fixed.

CHALLENGES AND FUTURE PERSPECTIVES FOR NITINOL ALLOYS IN BIOMEDICAL IMPLANTS

Along with summarizing recent findings, it is imperative to consider the underlying scientific challenges and new approaches in the development of NiTi shape memory alloys to prevent prosthesis failures and Ni release problems. Even with great advancements, some challenges still exist:

- Despite the excellent features of NiTi shape memory alloys, its nickel content raises concerns related to cytotoxic and allergic reactions in patients, which can lead to transplant rejection. This is a significant problem.
- It is sometimes difficult to precisely adjust the transformation temperature, which is necessary for NiTi to exhibit its characteristics, such as the shape memory effect and superelasticity.
- Longer research is needed to assess these alloys' long-term biological performance, particularly in the ion-rich human environment.

- A significant issue for NiTi in medical applications is its unique characteristics, which make it challenging to mill using traditional machining techniques. As a result, poor surface integrity increases the danger of Ni^+ ion release in the human biological system.[72] Although there are other non-traditional machining techniques that offer superior surface properties free from mechanical stresses, they nonetheless have certain drawbacks, such as heat-affected zones and recast layers. The machining properties of NiTi alloys, as well as their shape memory effect and shape recovery features, are the subject of a substantial research interest.[72]
- A coating on the NiTi surface can be developed in several ways. The longevity of the coating is an important issue that may be of interest to researchers in the future because of the frequent changes in the forms of SMA alloy.[72]
- Furthermore, there are several issues peculiar to the manufacturing and thermal treatment of NiTi alloy for shape, as it needs precise conditions to avoid losing its smart properties.

This review study provides several promising directions for further investigation to enhance the properties of NiTi alloys:

- Future research should concentrate on specific areas, such as the "long-term performance of surface-modified NiTi implants" or "developing novel manufacturing techniques to further enhance nitinol's properties."
- Working to develop surface modification and coating techniques to reduce nickel ions release and improve biocompatibility.
- Finally, applying artificial intelligence to forecast and enhance NiTi alloy behavior under challenging circumstances.[272]

CONCLUSION

This comprehensive review elucidates the distinctive properties of NiTi SMAs, emphasizing its corrosion resistance, tribological behavior, and the associated challenges and solutions concerning its biocompatibility. The principal conclusions drawn from this study are as follows:

- The unique shape memory effect, superelasticity, and biocompatibility make NiTi SMAs a pivotal material in advancing biomedical applications.
- The passive layer of NiTi SMAs, chiefly composed of TiO_2, provides exceptional resistance to wear and corrosion
- Ensuring successful application in medicine requires meticulous management throughout the production process, as defects can persist in the final product.
- The hardness values of NiTi SMAs ranged from 201 to 707 HV, indicating a significant improvement in strength.
- The coefficient of friction of NiTi alloys reduced to 0.31, which is lower compared to some alloys based on Ti, indicating improved wear resistance.
- It is considered that NiTi alloys are not cytotoxic if the cell viability remains $> 70\%$.
- Reducing wear and friction is essential to lessening the negative biological reactions associated with the production of debris.
- Extensive research on surface modifications has further augmented NiTi's biocompatibility and antibacterial properties, enhancing its stature as a preferred material for various medical devices.

AUTHOR CONTRIBUTIONS

N.B: Writing-original draft, visualization, resources, data curation. M.F.: Supervision, data curation, writing-original draft preparation, conceptualization, methodology, software, reviewing, and editing. H.N.: writing-original draft preparation. A.G.: Data curation. L.K: Funding, Reviewing and editing.

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DATA AVAILABILITY

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

CONFLICT OF INTERESTS

The authors have declared no competing financial interests or personal relationships that could have influenced the work reported in this paper.

ETHICAL APPROVAL

Not applicable.

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